Title: *The relationship of how many minutes someone spends sitting and how much difficulty they have with walking, running, or playing.*

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Summary:

Often in physical fitness culture there is fearmongering surrounding sedentary activity. Influencers often claim that spending too much time sitting and not enough time moving around or standing. Apple Watches even have automated alerts to make sure that you stand for at least one minute during 12 different hours of the day. I was curious to see if there was a noticeable association between sedentary activity and mobility limits.

The goal can be surmised in this question: Is there a link between how many minutes someone aged 3-19 spends sitting and how much difficulty they have with crawling, walking, running and/or playing?

Samples were taken from the 2012 The National Health and Nutrition Examination Survey (NHANES). In order to compare two independent samples with unequal variances, a Welch's t-test was used. This t-test accounts for the different variances because it does not pool the variances as it happens in Student's t-test.

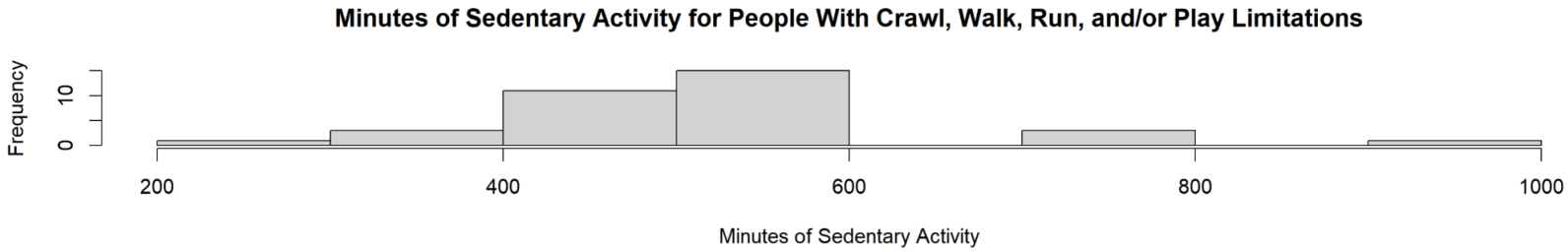
It was concluded that there is insufficient evidence that there is a significant difference in the average number of minutes of sedentary activity between individuals with and without mobility limitations.

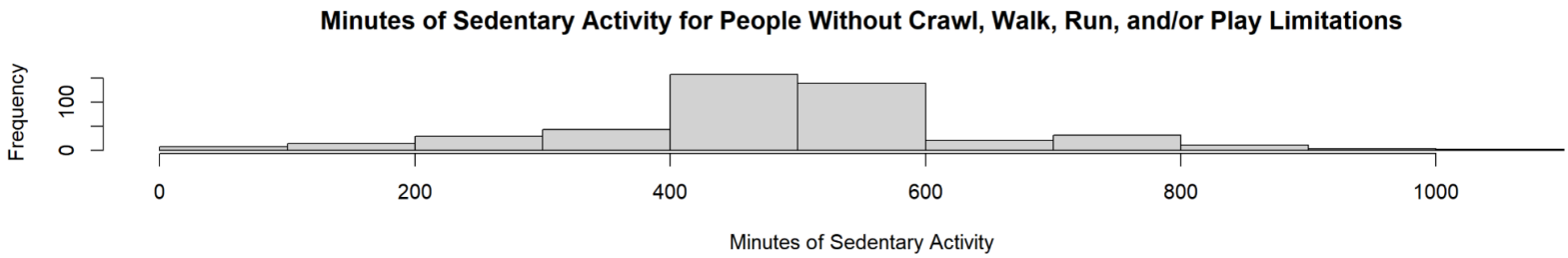
Introduction:

There are many sources that claim that people need to be standing and moving throughout the day to maintain their mobility. Many claim that if we do not move throughout the day, overtime, there would be consequences in the form of lower freedom of movement. However, I have seen little data backing these claims. I was curious to see if I could find a positive correlation between time spent sitting and experiencing limited mobility. This information would be helpful to many different people – given that there is a relationship between the two, those who are on the fence about decreasing their sedentary time could be pushed to pursue a more active lifestyle. It is even possible that people without any interest in becoming more active would be persuaded to begin considering it or even to begin to do so. It is also possible that there is no significant relationship between the sedentary activity and crawling, walking, running, and/or playing limitations. In this case, perhaps those that were on the fence would have their minds at ease that their lifestyle is not as damaging as it has sometimes been made out to be. It may also happen that someone lives a lifestyle that makes it difficult to spend time moving around. Individuals who work jobs that require them to sit at a desk for the majority of their days may find it hard to take some time to move each hour. These results are important because they could either reinforce people's motivation to move more and benefit their health, or ease the minds of those who struggle to move or work in a desk job. Ignorance is a source of confusion and distress. Life has its challenges for everyone, and mindfulness of one's mental and physical health is becoming increasingly more prevalent, but there remains much uncertainty of the ideal lifestyle.

In order to explore whether there is statistical evidence backing this claim, this analysis aims to answer the question "Is there a link between how many minutes someone aged 3-19 spends sitting and how much difficulty they have with walking, running, or playing?".

This analysis will use samples taken from the 2012 The National Health and Nutrition Examination Survey (NHANES). The Physical Activity and Physical Functioning surveys were used to assess how many minutes surveyed individuals spent sitting, and if they had difficulty crawling, walking, running, or playing. After cleaning the data using deletion to account for the many missing entries in minutes of sedentary data, there were 493 samples remaining. After separating those samples into groups, the number of those without movement limits is 34, and 457. Since the sample sizes for those that have difficulties moving and those that do not are vastly different, it is important that the test utilize an unpooled variance value. To compare two samples with unequal variances, Welch's t-test is the most appropriate method to use. This test assumes that the populations of both samples are normally distributed, which can be checked using a histogram (shown below).





Exploratory Data Analysis:

As aforementioned, The Physical Activity and Physical Functioning surveys were taken from The National Health and Nutrition Examination Survey (NHANES) of 2012. The variable “Crawl, run, walk, play limitations” was taken from the Physical Functioning survey. The variable is categorical, using numerical values to represent answers to the question “{Do you/Does SP} have an impairment or health problem that limits {your/his/her} ability to {walk, run or play} {walk or run}?“. It should be noted that individuals aged 16-19 were not questioned on their limits in playing. The categories for the variables are as follows: 1, Yes; 2, No; 7, Refused; 9, Don’t Know. From the Physical Functioning survey, “Minutes sedentary activity” was extracted. This is a continuous numerical variable that describes how survey responders answered the question “How much time {do you/does SP} usually spend sitting on a typical day?” (NHANES,2012). It should also be noted that both surveys questioned the same people, but the target ages for these specific questions were 3-15 and 12-15 respectively. It is likely that the smaller number of samples with entries for minutes spent sedentary is due to a non-response bias that results from their interest in a smaller range of ages. It could be that they did not survey individuals outside this age range. R was used to delete the missing entries of minutes spent sedentary and the related entries of mobility limitations in the effort of cleaning the data. Deletion was deemed to be the most appropriate, as each responder’s answer is independent, making any insertion technique unfit for cleaning the missing data. After doing so, there were a total of 491 samples with the relevant data entered. Of these remaining samples of mobility limitations, the only respondent out of the 1640 respondents – which matches the total number of individuals surveyed, who answered 9 (Don’t Know) was filtered out. Since this represents an outlier, this is beneficial to cleaning the data. It also would not be helpful in answering the research question, and is thus irrelevant. If there were a corresponding value for minutes of sedentary activity, though, perhaps it could have been useful in acting as a bias check of some sort by providing a sample with no affiliation at all. This leaves a remaining 457 who answered No, compared to only 34 who said Yes. These sample sizes are very different, and have very different variances as a result. This will need to be considered when selecting an analysis approach.

To get more information on the minutes of sedentary activity, R was used to both summarize and describe the data set. The outputs of both functions are below, as labeled.

```
[1] "Summary:"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   10.0   420.0   480.0   502.5   600.0  1080.0
[1] "Description:"
minsSedentaryData
       n  missing distinct     Info     Mean  pMedian      Gmd      .05
     491        0       28    0.978    502.5      510    167.7      240
     .10      .25      .50      .75      .90      .95
     300      420      480      600      720      720

lowest :   10   30   60  120  150, highest:  780  840  900  960 1080
```

R was also then used to produce a histogram of the cleaned data of minutes spent sedentary. The data appears to be approximately normal – bimodal with no outliers.

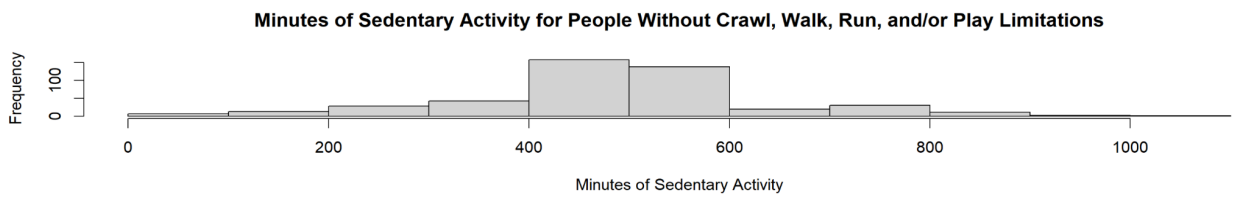


Figure 3

A scatterplot to assess independence was also done with R, shown below.

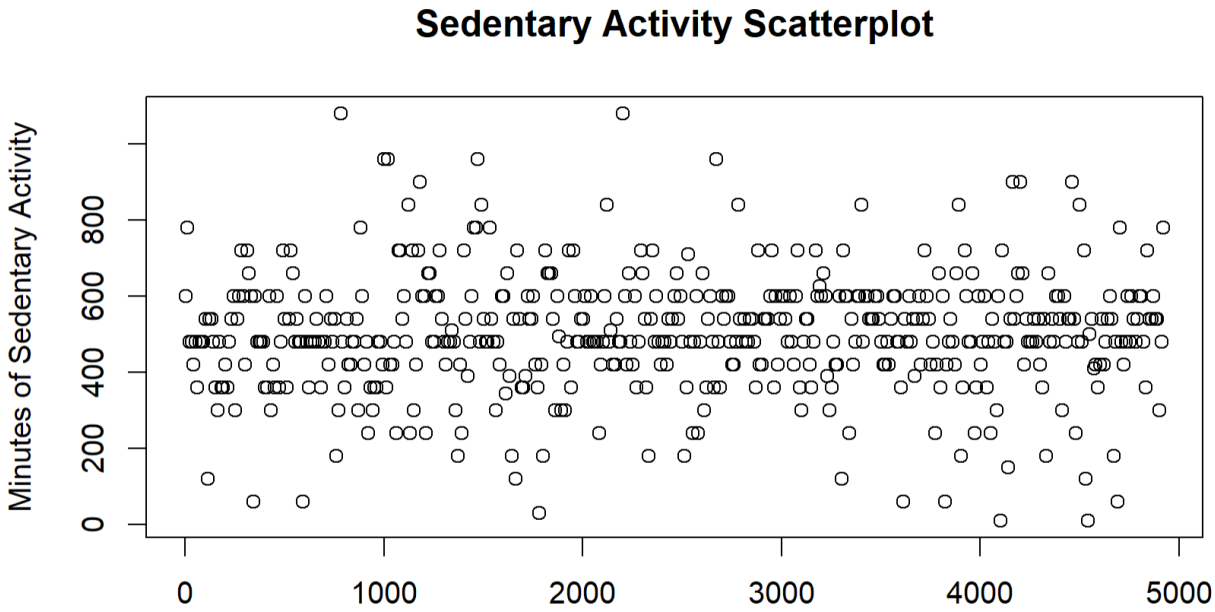


Figure 4

The data appears to be random, reinforcing the independence. The experiment design, which is assessing different individuals, also reinforces the idea of independence. Assuming random samples, one individual's time spent sitting should have no effect on another's.

I will proceed with the statistical analysis with these biases and considerations in mind.

Data Analysis:

In order to analyze two independent samples with unequal variances, Welch's t-test must be used. Student's two sample t-test requires equal variances and uses a pooled approximation of variance in its calculations. Welch's t-test, conversely, does not use the same approach, making it more robust and versatile, and increasing its utility in this analysis.

Welch's t-test does, however, still require that the samples be taken from a population with an approximately normal distribution. Normality has already been assessed during the exploratory analysis phase with the histograms (Figure 3), thus this condition is satisfied. With sample sizes of 457 and 34, it is safe to assume that these samples are less than 10 percent of their populations. Minutes of sedentary activity is a continuous variable, which is another condition that must be satisfied. The test also requires random samples, which was also assessed in the exploratory analysis (Figure 4).

All conditions are satisfied, and I will proceed with the analysis while still keeping the biases found in the exploratory analysis in mind.

The hypotheses for the Welch's t-test for this analysis are as follows:

Null Hypothesis: The average number of minutes spent sedentary is the same for individuals with and without mobility limits.

Alternative Hypothesis: The average number of minutes spent sedentary is different for individuals with and without mobility limits.

*NB: The following errors are possible with these hypotheses:

- Type I Error: Concluding that there is a significant difference when in reality there is not (null hypothesis unduly rejected)
- Type II Error: Concluding that there is not a significant difference when in reality there is (null hypothesis unduly accepted)

The chosen significance level was .05, as this is the significance level most commonly used in practice. The sample size is 491 – 457 with no limits and 34 with limits. All calculations were done using R's built in functions. The code for the test is shown below.

```
# PERFORMING WELCH TWO SAMPLE T-TEST

results <- t.test(minsSedentaryYesData, minsSedentaryNoData)
```

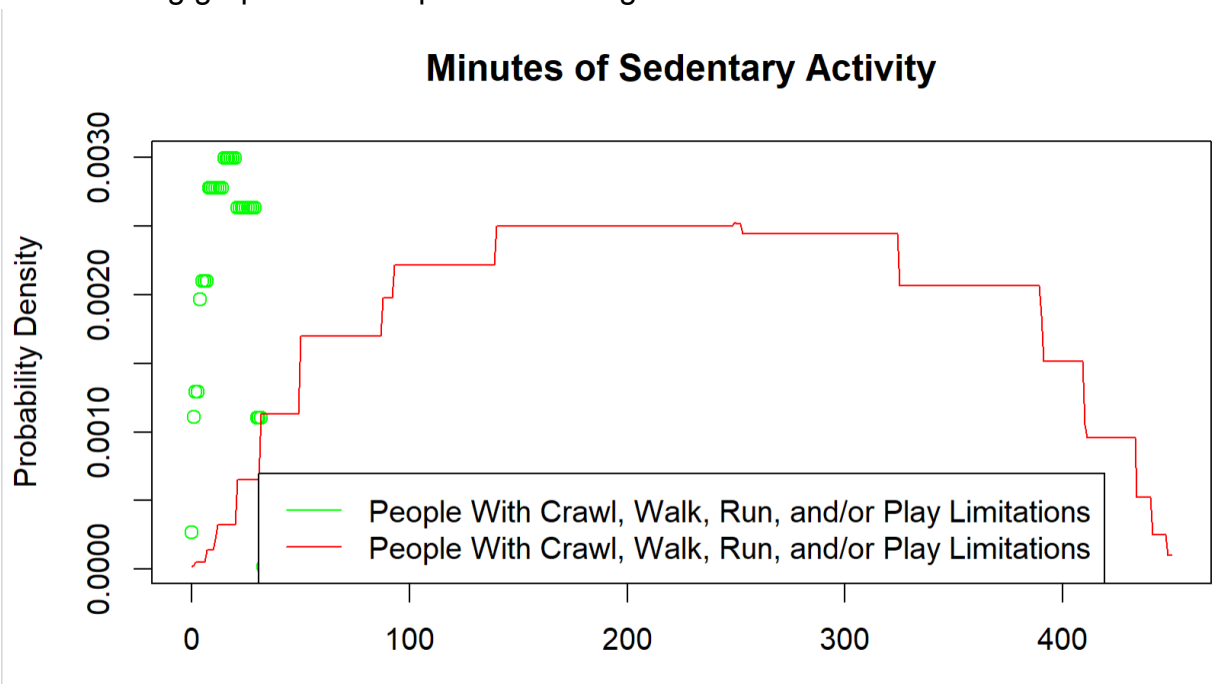
The output of the test is shown below.

```
> results

        Welch Two Sample t-test

data:  minsSedentaryYesData and minsSedentaryNoData
t = 1.3331, df = 40.29, p-value = 0.19
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -16.46364  80.31086
sample estimates:
mean of x mean of y
 532.2059  500.2823
```

Results:

With a p value of 0.19, there is insufficient evidence that there is a significant difference in the average number of minutes of sedentary activity between individuals aged 3-19 with and without mobility limitations with 95% confidence.

The following graph was also produced using R.



Looking at this graph, it does appear that there is a significant difference between the two means. However, there is reason to believe the vastly different variances had a large effect on the outcome of the test. It is also important to keep in mind the biases that were found in the exploratory analysis. It is also possible that people with extreme limitations in mobility were skipped when asking about their sedentary activity, due to

necessity resulting from their conditions. It is also important to keep in mind that this survey was limited to individuals aged 3-19, and that limitations resulting from sedentary activity might not appear until later in life.

Conclusion:

In conclusion, there is insufficient evidence that there is a significant difference in the average number of minutes of sedentary activity between individuals aged 3-19 with and without mobility limitations. Further analysis would have to be performed to determine if more extreme consequences appear long-term.

References:

Centers for Disease Control and Prevention. (n.d.). *NNYFS questionnaire data*. Centers for Disease Control and Prevention. https://wwwn.cdc.gov/nchs/nhanes/search/nnyfsdata.aspx?Component=Questionnaire